Fakultas Informatika
School of Computing
Telkom University

Aplikasi Berbasis Platform

Javascript & jQuery

Objectives

- Student understands the usage of Javascript in the web programming
- Student able to implement Javascript structure inside HTML document
- Student able to integrate HTML + CSS+ Javascript on a program

Javascript Overview

- Available on a browser at client side
- Used to make web pages interactive
  - Insert dynamic text/tag into HTML (ex: comment box)
  - **React to events** (ex: page load, user click, user input)
  - Perform logic on user's computer (ex: form validation, calculate total price in shopping cart)
- Javascript != Java
- NOT related to Java other than by name and some syntactic similarities

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Javascript Embed inside HTML

```
<!- some HTML code -->
<script type="text/javascript">
       // Javascript code here
</script>
<!- some HTML code -->
```

- Embedded Javascript can be placed anywhere inside tag <body> in HTML document

Linking to a external Javascript file


```
<script src="filename.js" type="text/javascript"></script>
```
HTML

- script tag usually placed in HTML page's <head> (like linked CSS) or on the very bottom of HTML document (before </html>)
- script code is stored in a separate file (xxx.js)

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Variables and Data Types

- Rules of naming methods
  - Case sensitive
  - Start with a letter or underscore
  - Must not a Javascript's reserved word
  - Must not contains operational characters

- Variable declaration:
  - var variableName = value;
  - variableName = value;

Variables


```
var name = expression;
```
JS

```
var clientName = "Connie Client";
var age = 32;
var weight = 127.4;
```
JS

- variables are declared with the *var* keyword (case sensitive) or *let*
- doesn't need to specify the type of variable, it's automatically adapt to the value within. But, JS has types ("loosely typed")
  - `Number, Boolean, String, Array, Object, Function, Null, Undefined`
  - can find out a variable's type by calling `typeof`

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Comments


```
// single-line comment
/* multi-line comment */
```
JS

- identical to Java's comment syntax
- recall: comment syntaxes
  - HTML: <!-- comment -->
  - CSS: /* comment */

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Number type


```
var enrollment = 99;
var medianGrade = 2.8;
var credits = 5 + 4 + (2 * 3);
```

JS

- integer and real are the same type (Number)
- same operators like C or Java: + - * / % ++ -- = += -= *= /= %=
- similar precedence to Java
- many operators auto-convert types: "2" * 3 is 6

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Boolean type


```
var iLike190M = true;
var ieIsGood = "IE6" > 0; // false
if ("web devevelopment is great") { /* true */ }
if (0) { /* false */ }
```

JS

- any value can be used as a Boolean
  - "falsey" values: 0, 0.0, NaN, "", null, and undefined
  - "truthy" values: anything else from falsey values
- converting a value into a Boolean explicitly:
  - `var boolValue = Boolean(otherValue);`
  - `var boolValue = !!(otherValue);`

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String type


```
var s = "Connie Client";
var name1 = s.substring(2, 8); // "nnie C"
var name2 = s.substr(2, 8); // "nnie Cli"
var len = s.length; // 13
var ie1 = s.indexOf('ie'); // 4
var ie2 = s.lastIndexOf('ie'); // 9
var firstName = s.substring(0, s.indexOf(" ")); //Conni
```

JS

- other methods: `charAt, replace, split, join, reverse, toLowerCase, toUpperCase`
  - charAt returns a one-letter String (there is no char type)
- length is a property (not a method as in Java)
- strings can be specified with double quote ("") or single quote ('')
- concatenation with + :
  - 1 + 1 is 2, but "1" + 1 is "11"

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More about `String`

- escape sequences behave as in Java:

\' \" \& \n \t \\

- converting between numbers and Strings:

```
var count = 10;
var s1 = "" + count; // "10"
var s2 = count + " bananas, full!"; // "10 bananas, full!"
var n1 = parseInt("42 is the answer"); // 42
var n2 = parseFloat("booyah"); // NaN                    JS
```

- accessing the letters of a String:

```
var firstLetter = s[0]; // fails in IE
var firstLetter = s.charAt(0); // does work in IE
var lastLetter = s.charAt(s.length - 1);                JS
```

Splitting strings: split and join


```
var s = "the quick brown fox";
var a = s.split(" "); // ["the", "quick", "brown", "fox"]
a.reverse(); // ["fox", "brown", "quick", "the"]
s = a.join("!"); // "fox!brown!quick!the"
```

JS

- split breaks apart a string into an array using a delimiter
  - can also be used with regular expressions (seen later)
- join merges an array into a single string, placing a delimiter between them

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Arrays


```
var name = []; // empty array
var name = [value, value, ..., value]; // pre-filled
name[index] = value; // store element
```

JS

```
var ducks = ["Huey", "Dewey", "Louie"];
var len = ducks.length // 3
var stooges = []; // stooges.length is 0
stooges[0] = "Larry"; // stooges.length is 1
stooges[1] = "Moe"; // stooges.length is 2
stooges[4] = "Curly"; // stooges.length is 5
stooges[4] = "Shemp"; // stooges.length is 5
```

JS

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Array methods


```
var a = ["Stef", "Jason"]; // Stef, Jason
a.push("Brian"); // Stef, Jason, Brian
a.unshift("Kelly"); // Kelly, Stef, Jason, Brian
a.pop(); // Kelly, Stef, Jason
a.shift(); // Stef, Jason
a.sort(); // Jason, Stef
```

JS

- array serves as many data structures: list, queue, stack, ...
- methods: `concat, join, pop, push, reverse, shift, slice, sort, splice, toString, unshift`
  - push and pop add / remove from back
  - unshift and shift add / remove from front
  - shift and pop return the element that is removed

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Logical operators

- > < >= <= && || ! == != === !==
- most logical operators automatically convert types:
  - 5 < "7" is true
  - 42 == 42.0 is true
  - "5.0" == 5 is true
- === and !== are strict equality tests; checks both type and value
  - "5.0" === 5 is false

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if/else statement (same as Java)

```
if (condition) {
        statements;
} else if (condition) {
        statements;
} else {
        statements;
}
```

JS

- identical structure to Java's if/else statement
- Javascript allows almost anything as a condition

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for loop

```
for (var i = 1; i < 7; i++) {
  document.write("<h"+i+">Header "+i+"</h"+i+">");
}
// prints six HTML headers
```

JS

```
var s1 = "hello";
var s2 = "";
for (var i = 0; i < s.length; i++) {
      s2 += s1.charAt(i) + s1.charAt(i);
}
// s2 stores "hheellllloo"
```

JS

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while loops


```
while (condition) {
       statements;
}
```
JS

```
do {
    statements;
} while (condition);
```
JS

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Object

- Every element (HTML tag) in HTML document is a Javascript object (DOM object)

- Object types
  - Internal (built-in): String, Array, Math, Date
  - External: All HTML elements in HTML document, Window, user-defined object

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DOM element objects

HTML

```
<p>
  Look at this octopus:
  <img src="octopus.jpg" alt="an octopus" id="icon01" />
  Cute, huh?
</p>
```

DOM Element Object

Property	Value
tagName	"IMG"
src	"octopus.jpg"
alt	"an octopus"
id	"icon01"

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Math object


```
var rand1to10 = Math.floor(Math.random() * 10 + 1);
var three = Math.floor(Math.PI);
```
JS

- methods: `abs, ceil, cos, floor, log, max, min, pow, random, round, sin, sqrt, tan`
- properties: `E, PI`

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Window object

- Document
  - All HTML tags in HTML document is stored and can be accessed with this object
- History
- Location
- Pop-up boxes (Alert, Confirm, Prompt)

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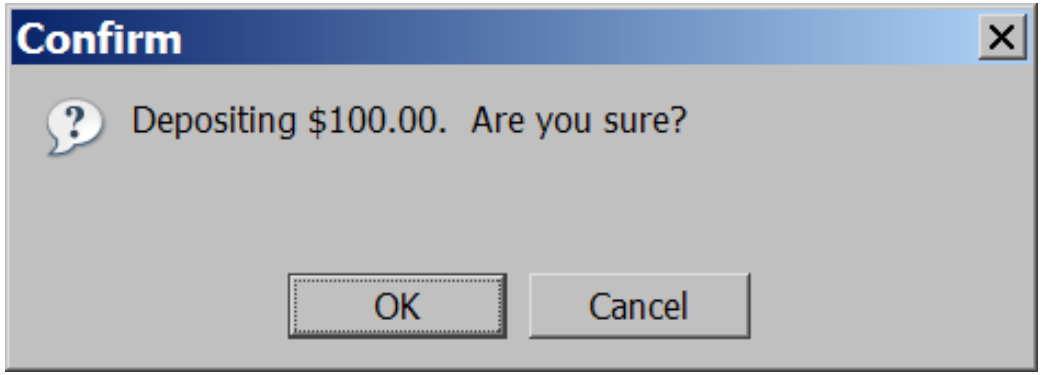
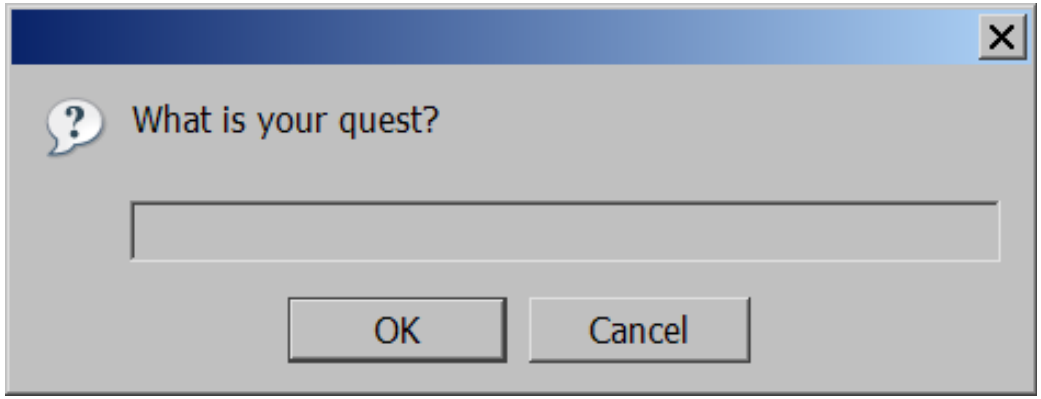
Popup boxes


```
alert("message"); // message
confirm("message"); // returns true or false
prompt("message"); // returns user input string
```

JS





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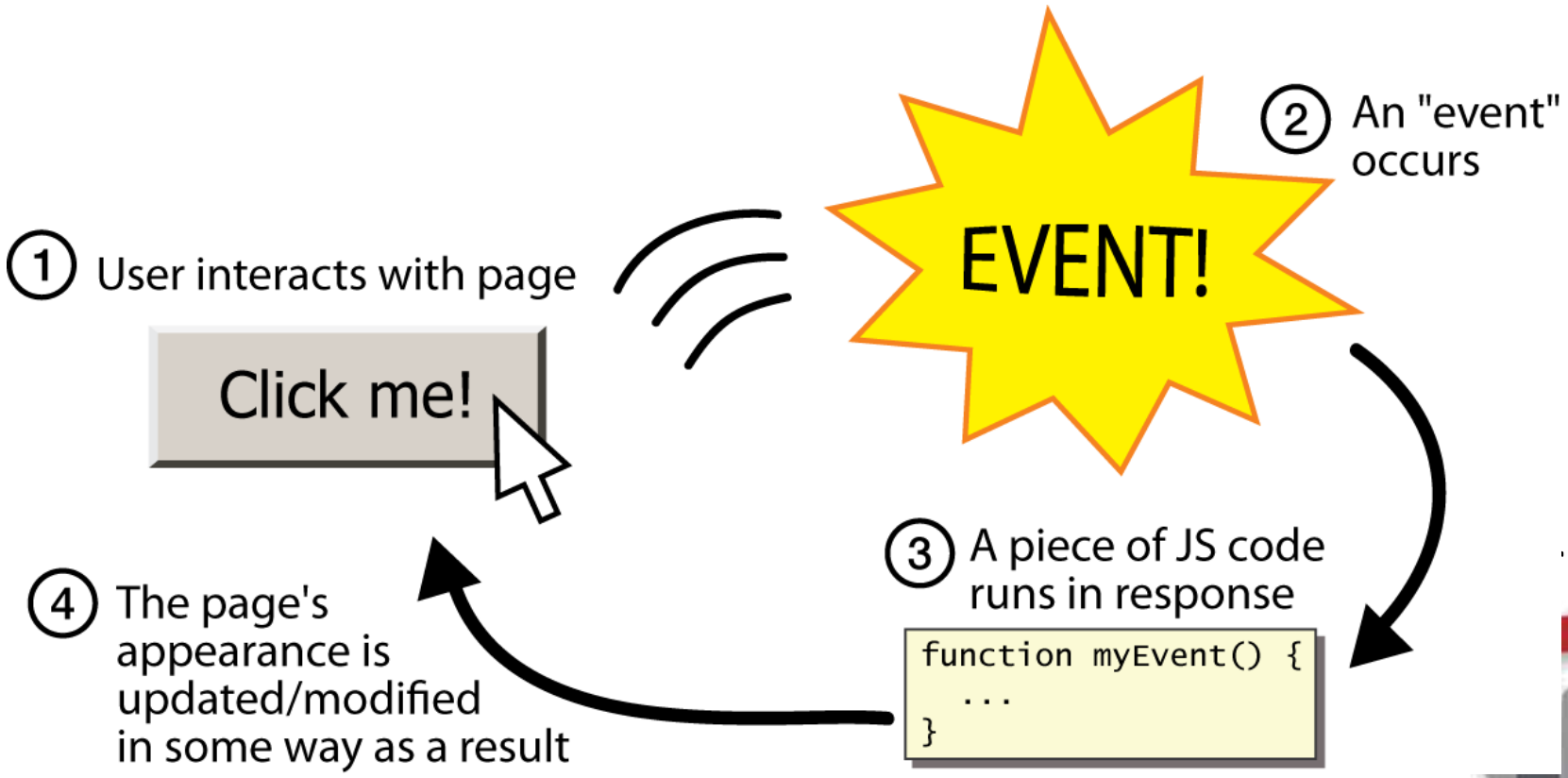
Event-driven programming



Javascript functions


```
function name() {
        statement ;
        statement ;
        ...
        statement ;
}
```
JS

```
//example
function mySalute() {
        alert("Hello!");
        alert("How are you?");
}
```
JS

- function usually is stored in the contents of external JS file linked to our HTML page
- statements placed into functions can be evaluated in response to user events

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Javascript functions


```
//function declaration
function add(bil1, bil2) {
       var result= bil1 + bil2;
       return result;
}
```
JS

```
//function execution
mySalute(); //no return value
var val = add(3,7); //with return value, assign variable
document.write(add(3,7)); /* with return value, as input
for another function */
```
JS

- What will be printed on the screen?

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Event handlers


```
<[element] ... on[event]="myFunction()"> ...
```
HTML

```
<button onclick="myFunction()">Click me!</button>
```
HTML

- Function in Javascript usually is used for event handler
  - when we interact with a tag (element), function will be executed
- To able a tag to do the event handler, certain attribute must be added to that tag, according to what event that tag want to capture. The attribute for event handler starts with "on.."
- `onclick` is just one of attribute for event handler. Code example above means: "if this button tag is **clicked**, then execute myFunction()"
- Another event handler attributes: `onchange, onfocus, onblur, onmouseover, onkeypress, etc`

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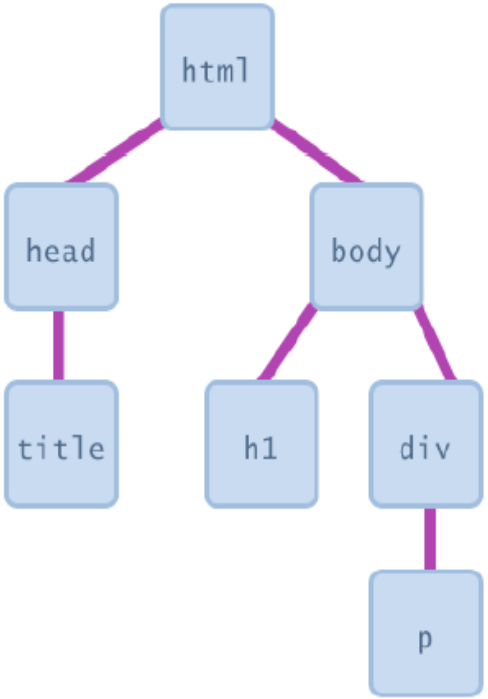
Document Object Model (DOM)

- JS can manipulates elements on HTML document
- we can examine elements' state
  - e.g. see whether a box is checked
- we can change state
  - e.g. insert some new text into a div, or add new div (box) into another div
- we can change styles
  - e.g. make a paragraph red



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DOM element objects

HTML

```
<p>
  Look at this octopus:
  <img src="octopus.jpg" alt="an octopus" id="icon01" />
  Cute, huh?
</p>
```

DOM Element Object

Property	Value
tagName	"IMG"
src	"octopus.jpg"
alt	"an octopus"
id	"icon01"

JavaScript

```
var icon = document.getElementById("icon01");
icon.src = "kitty.gif";
```

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Accessing object (HTML element)

- function *document.getElementById("x")* returns the DOM object from the element which has the *id* attribute value = "x"
  - we can get or change the text in <input> element by setting the *value* attribute
  - we can get or change the text inside most elements by setting the *innerHTML* attribute

Accessing object (HTML element)

- Example

```
// returns whole DOM object of element with id = “idname”
var obj = document.getElementById("idname");

// returns only the content of element with id = “idname”
// ex 1: get the content from element <input />
var val = document.getElementById("idname").value;
// ex 2: get the content from element other than <input />
var con = document.getElementById("idname").innerHTML;
```

JS

Manipulating element


```html
<input id="textbox" type="text" />
<div id="output">replace me</div>
<button onclick="getValue()">Click me!</button>
<button onclick="changeText()">Click me too!</button>
```

HTML

```js
function getValue() { //get the content of element
  var text = document.getElementById("textbox").value;
  var con = document.getElementById("output").innerHTML;
  alert("value: " + text + " & " + con);
}
function changeText() { //manipulating element
  //method 1: indirect, from returning object
  var sp1 = document.getElementById("output");
  sp1.innerHTML = "replaced! <br /> <button>New</button>";
  //method 2: direct assignment on object attribute
  document.getElementById("textbox").value = "Connie";
}
```

JS

Any question?

jQuery

What is jQuery?

- jQuery is a lightweight, "**write less, do more**", JavaScript framework.
- jQuery was originally released in January 2006 at BarCamp NYC by **John Resig** and was influenced by Dean Edwards' earlier cssQuery library
- The purpose of jQuery is to make it much easier to use JavaScript on your website.
- jQuery takes a lot of common tasks that require many lines of JavaScript code to accomplish, and wraps them into methods that you can call with a single line of code.
- There are lots of other JavaScript frameworks out there, but jQuery seems to be the most popular, and also the most extendable.

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Adding jQuery to Your Web Pages

- Download the jQuery library from jQuery.com

```
<head>
      <script src="jquery-3.6.1.min.js"></script>
</head>
```

- or include jQuery from external URL (you must always connect to internet)

```
<head>
      <script src="https://code.jquery.com/jquery-
3.6.1.min.js"></script>
</head>
```

jQuery Syntax

- Basic syntax is: **$(*selector*).*action*()**
  - A $ sign is to define/access jQuery
  - A (*selector*) is to "query (or find)" HTML elements (like CSS selector)
  - An *action*() is a jQuery function to be performed on the element(s)
- Examples:
  - $("p").hide() - hides all <p> elements.
  - $(".test").hide() - hides all elements with class="test".
  - $("#test").hide() - hides the element with id="test".

The Document Ready Event

- You might have noticed that all jQuery methods are inside a document ready event:
- This is to prevent any jQuery code from running before the document is finished loading (is ready).
- It is good practice to wait for the document to be fully loaded and ready before working with it. This also allows you to have your JavaScript code before the body of your document, in the head section.

```
$(document).ready(function(){
   // jQuery methods go here...
});

$(function(){
   // jQuery methods go here...
});
```

jQuery Selectors

Syntax	Description
$("*")	Selects all elements
$(this)	Selects the current HTML element
$("p.intro")	Selects all <p> elements with class="intro"
$("p:first")	Selects the first <p> element
$("ul li:first")	Selects the first <li> element of the first <ul>
$("ul li:first-child")	Selects the first <li> element of every <ul>
$("[href]")	Selects all elements with an href attribute
$("a[target='_blank']")	Selects all <a> elements with a target attribute value equal to "_blank"
$("a[target!='_blank']")	Selects all <a> elements with a target attribute value NOT equal to "_blank"
$(":button")	Selects all <button> elements and <input> elements of type="button"
$("tr:even")	Selects all even <tr> elements
$("tr:odd")	Selects all odd <tr> elements

jQuery Event Methods

- click, keypress, submit, load, dblclick, keydown, change, resize, mouseenter, keyup, focus, scroll, mouseleave, blur, unload

```
$(document).ready(function(){
   $("p").click(function(){
      // action goes here!!
   });
});
```

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jQuery - Get Content and Attributes

- Three simple, but useful, jQuery methods for DOM manipulation are:
  - text() - Sets or returns the text content of selected elements
  - html() - Sets or returns the content of selected elements (including HTML markup)
  - val() - Sets or returns the value of form fields
  - ```
    $("#btn2").click(function(){
        alert("HTML: " + $("#test").html());
    });
    ```
- The jQuery attr() method is used to get attribute values.
  - ```
    $("button").click(function(){
        alert($("#w3s").attr("href"));
    });
    ```
- The jQuery data() method is used to get attribute values.
  - ```
    $("button").click(function(){
        alert($("#w3s").data("option"));
    });
    ```

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    ```

jQuery - Set Content and Attributes

- We will use the same three methods from the previous page to **set content**:
  - text() - Sets or returns the text content of selected elements
  - html() - Sets or returns the content of selected elements (including HTML markup)
  - val() - Sets or returns the value of form fields
  - ```
    $("#btn2").click(function(){
        $("#test2").html("Hello world!");
    });
    ```
- The jQuery attr() method is also used to set/change attribute values.
  - ```
    $("button").click(function(){
        $("#w3s").attr({
            "href" : "https://www.w3schools.com/jquery/",
            "title" : "W3Schools jQuery Tutorial"
        });
    });
    ```

jQuery - Add Elements

- We will look at four jQuery methods that are used to add new content:
  - append() - Inserts content **at the end** of the selected elements
    - `$("table").append("<tr><td>add row append table</td></tr>");`
  - prepend() - Inserts content **at the beginning** of the selected elements
    - `$("table").append("<tr><td>add row prepend table</td></tr>");`
  - after() - Inserts content **after the selected** elements
    - `$("p").after("<span>Some text after</span>");`
  - before() - Inserts content **before the selected** elements
    - `$("p").after("<span>Some text before</span>");`

jQuery - Remove Elements

- remove() - Removes the selected element (and its child elements)
  - `$("#div1").remove();`
- empty() - Removes the child elements from the selected element
  - `$("#div1").empty();`
- The following example removes all <p> elements with class="test" and class="demo":
  - `$("p").remove(".test, .demo");`

jQuery - Get and Set CSS Classes

- Query has several methods for CSS manipulation. We will look at the following methods:
- addClass() - Adds one or more classes to the selected elements
  - `$("h1, h2, p").addClass("blue");`
- removeClass() - Removes one or more classes from the selected elements
  - `$("h1, h2, p").removeClass("blue");`
- toggleClass() - Toggles between adding/removing classes from the selected elements
  - `$("h1, h2, p").toggleClass("blue");`
- css() - Sets or returns the style attribute
  - `$("p").css("background-color", "yellow");`

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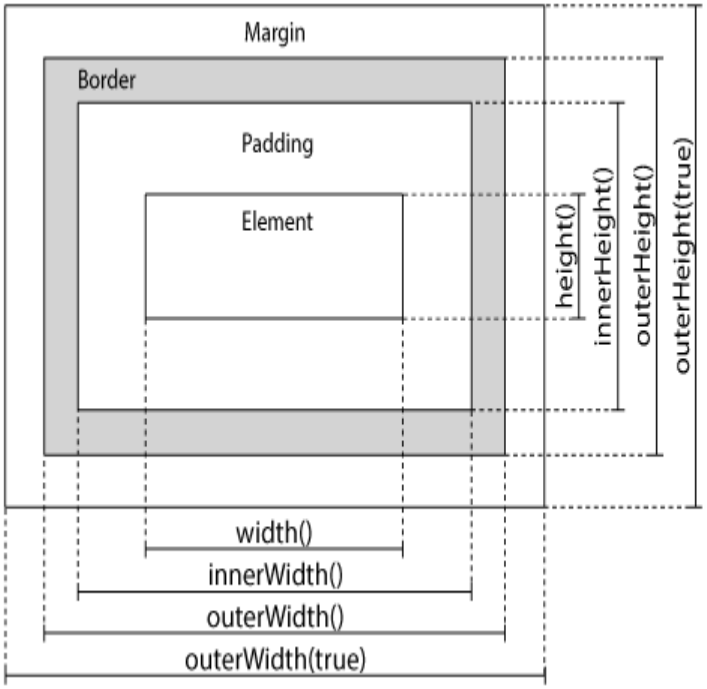
jQuery Dimension Methods

- jQuery has several important methods for working with dimensions:
  - width()
  - height()
  - innerWidth()
  - innerHeight()
  - outerWidth()
  - outerHeight()




```
$("button").click(function(){
    var txt = "";
    txt += "Width: " + $("#div1").w
    txt += "Height: " + $("#div1").
    $("#div1").html(txt);
});
```

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Looping over the DOM

```
<ul>
  <li>foo</li>
  <li>bar</li>
</ul>

$( "li" ).each(function() {
  $( this ).addClass( "foo" );
});
```

Any question?

Exercise

- Create a function that calculates only the odd values in a sequence of numbers, given the min and max sequence value as inputs. Example:
  - min = 1, max = 8, result = 16 (1+3+5+7)
  - min = 4, max = 15, result = 60 (5+7+9+11+13+15)
- Open your previous CSS exercise file, add two buttons that call a function when clicked:
  - First button: change the alignment of the text to center or right
  - Second button: change the background color of one of the box. When button is clicked again, the background color is back to original color

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References

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- http://www.webstepbook.com/supplements-2ed/slides/lectureXX-jquery.shtml
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